

PATENT A

Self-Sustaining Neural-Motor Energy Harvesting Loop■for Autonomous Cognitive Systems

8 drawing sheets

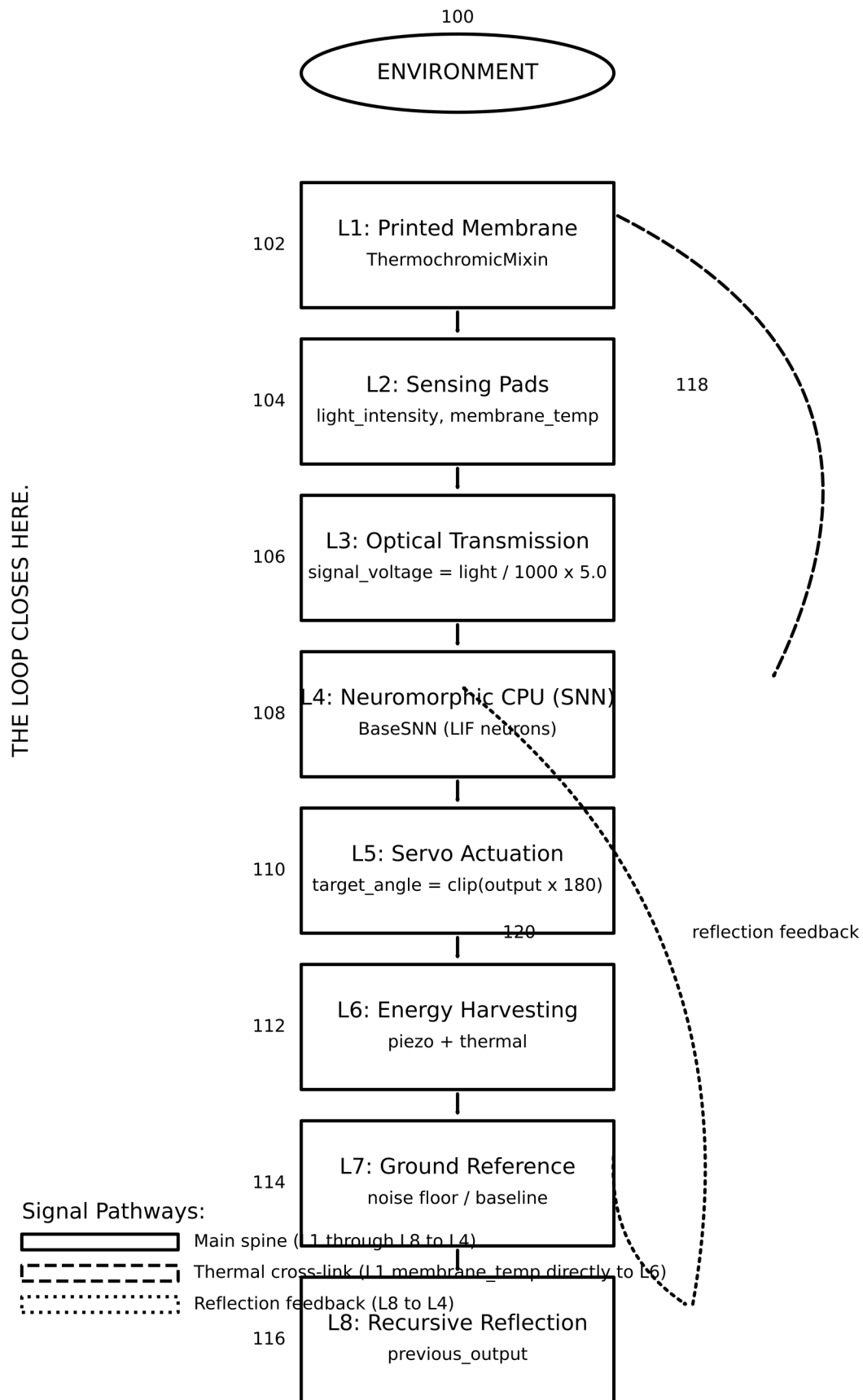


FIG. 1

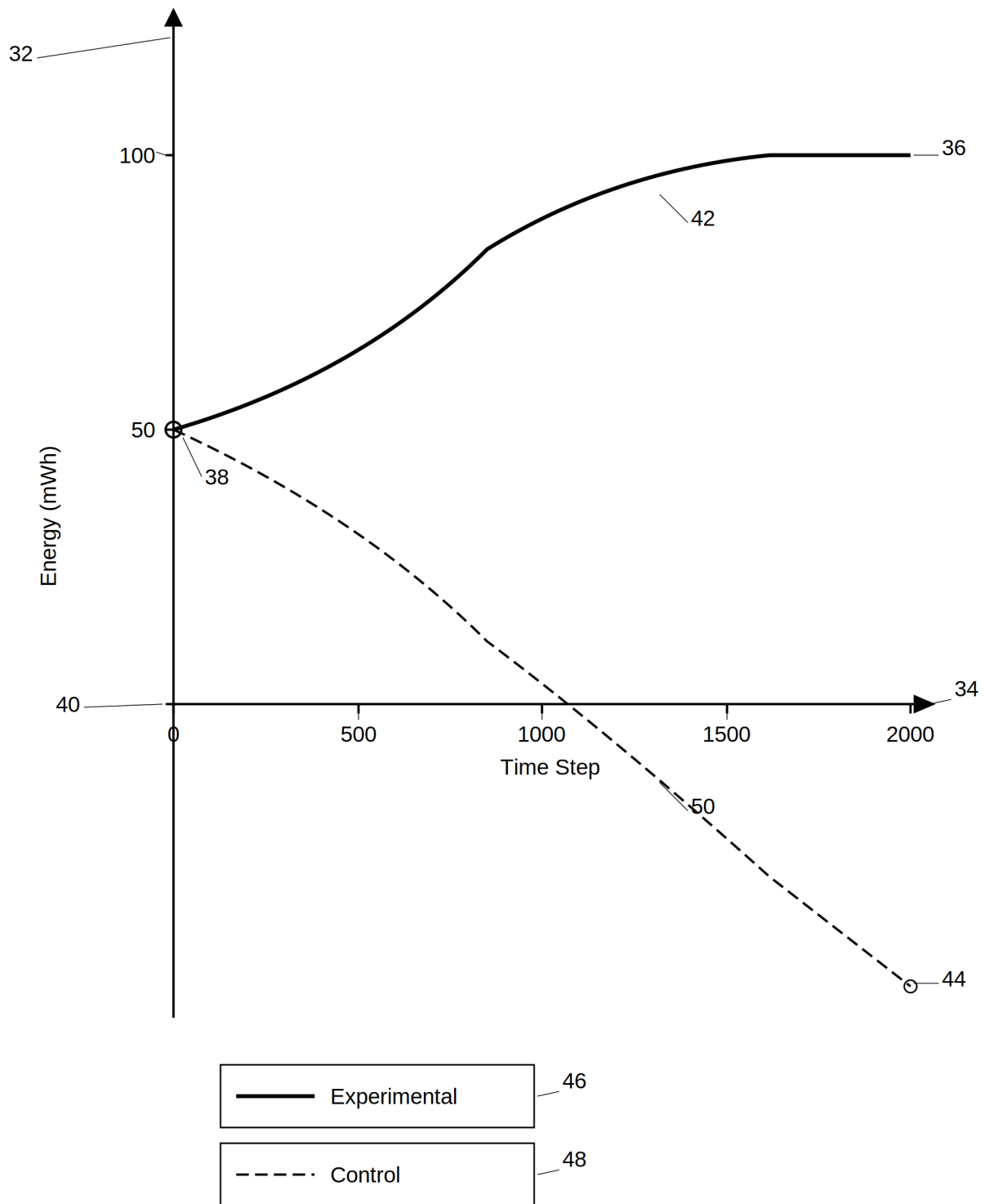


FIG. 2

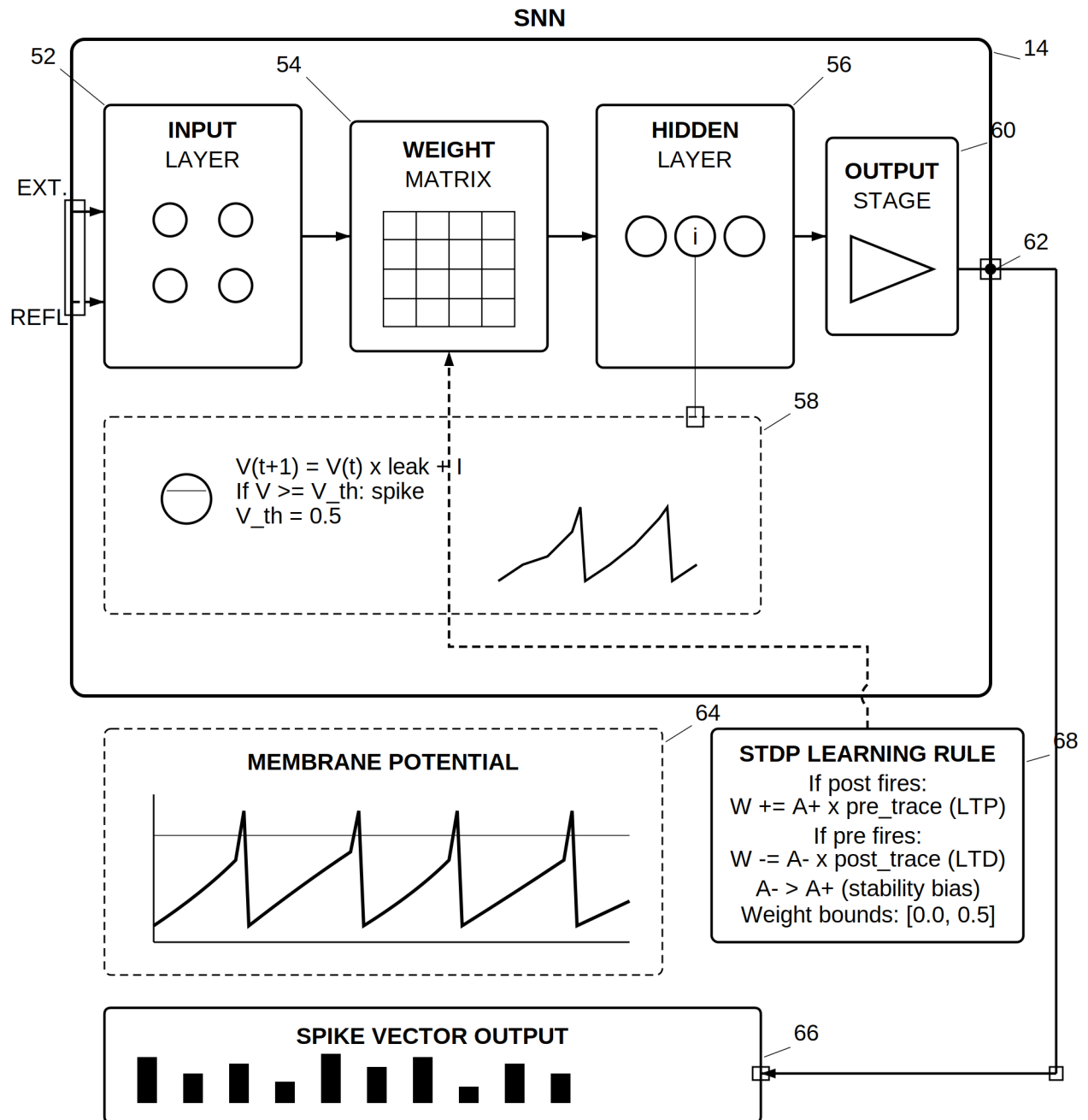


FIG. 3

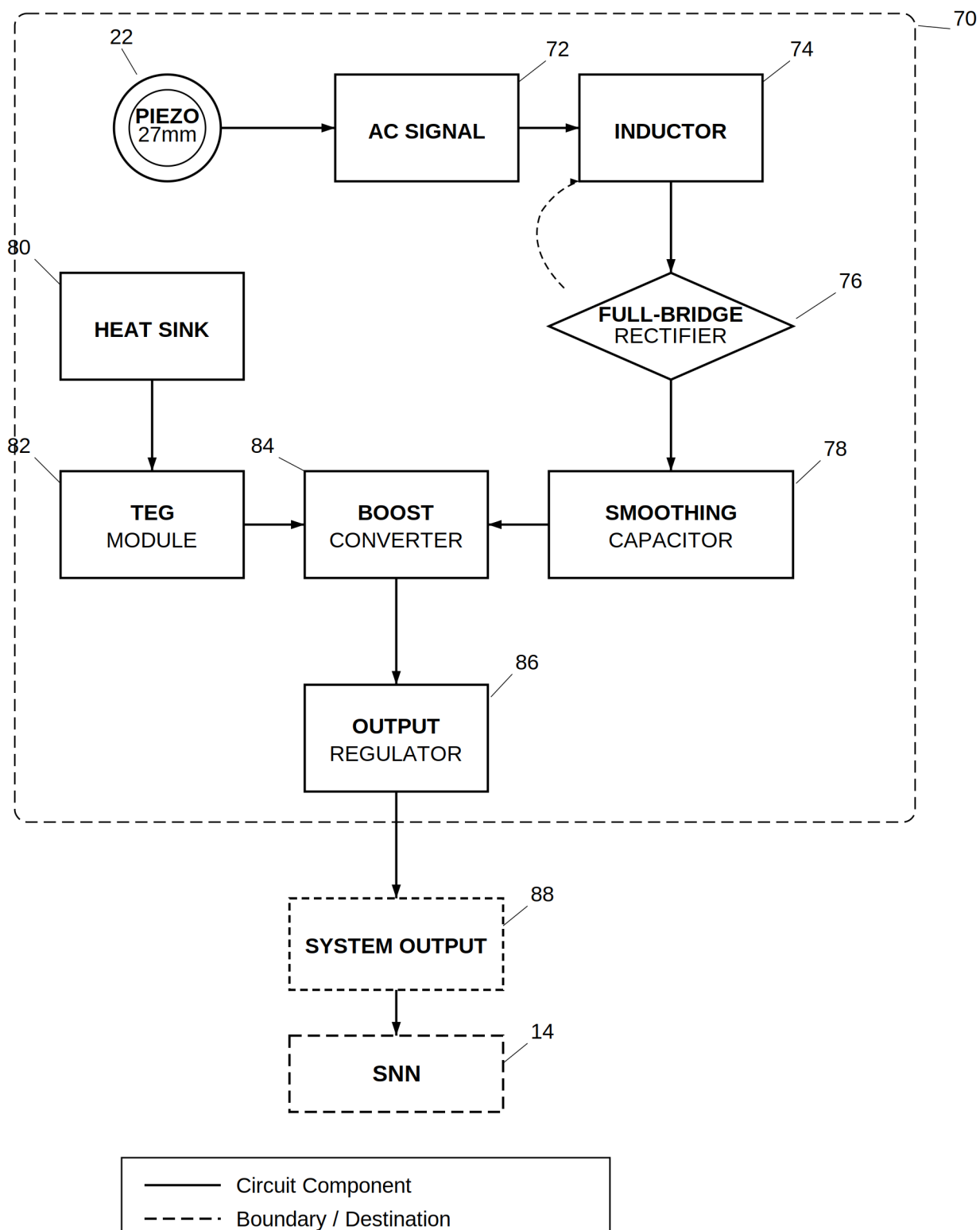


FIG. 4

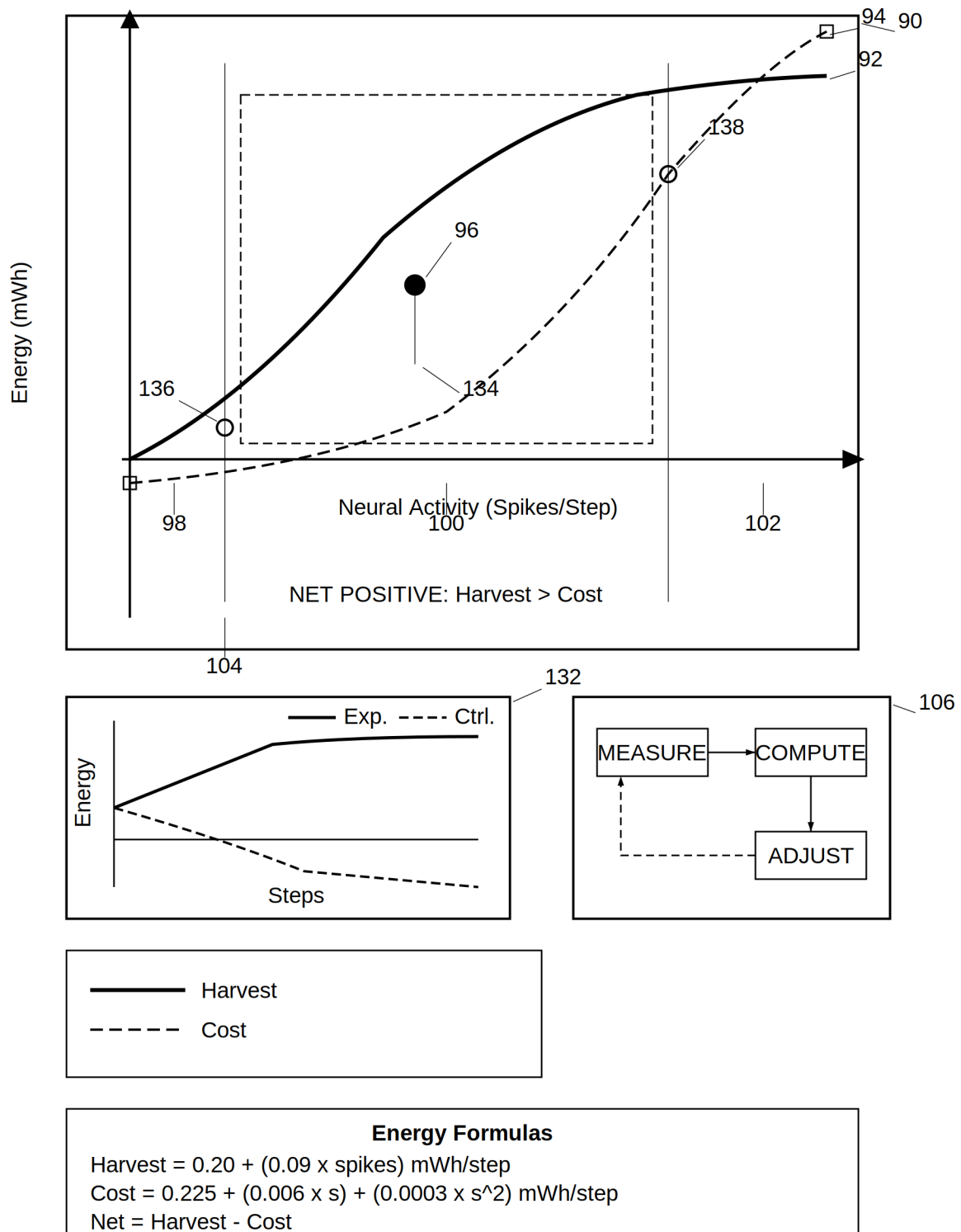


FIG. 5

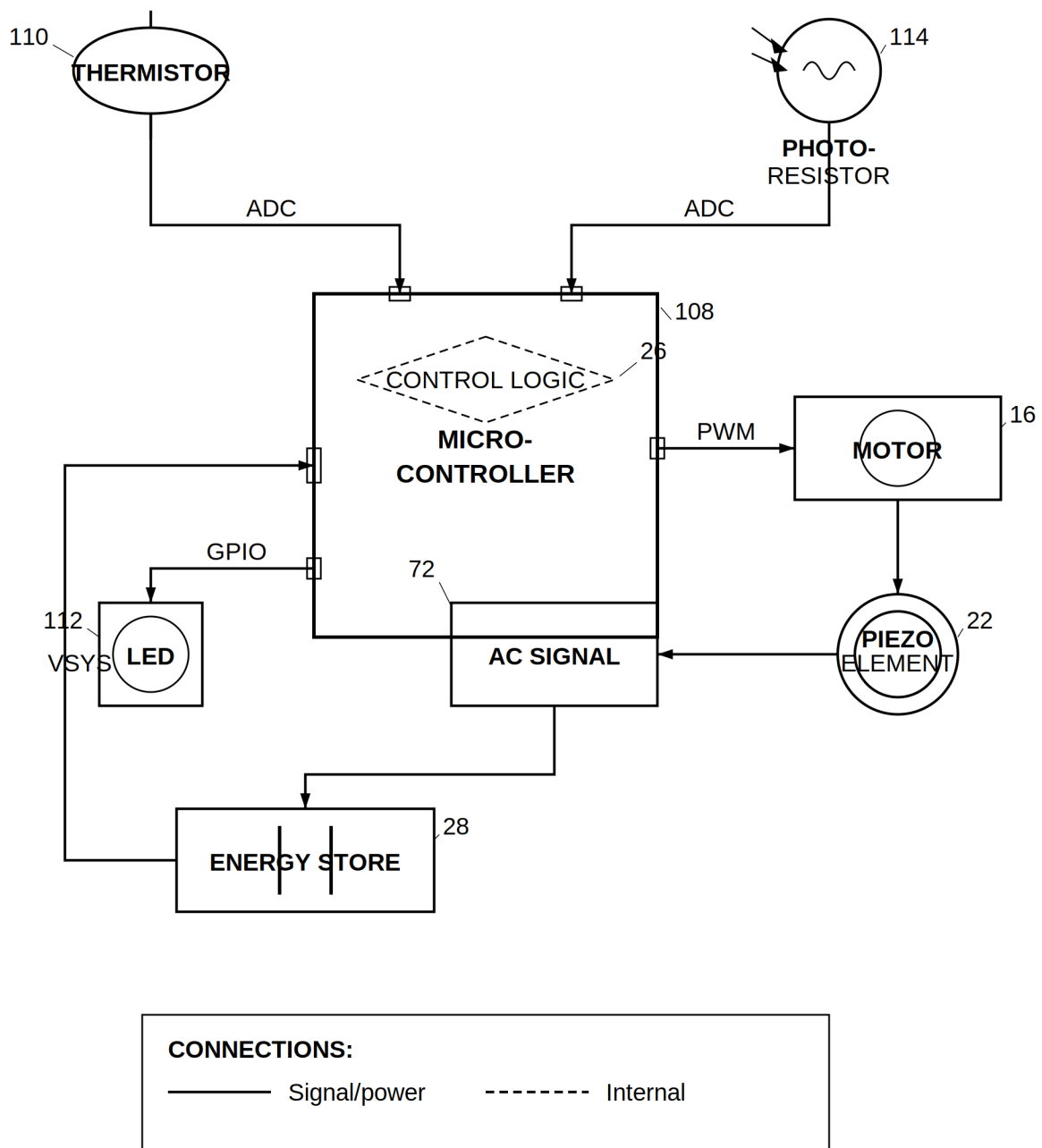


FIG. 6

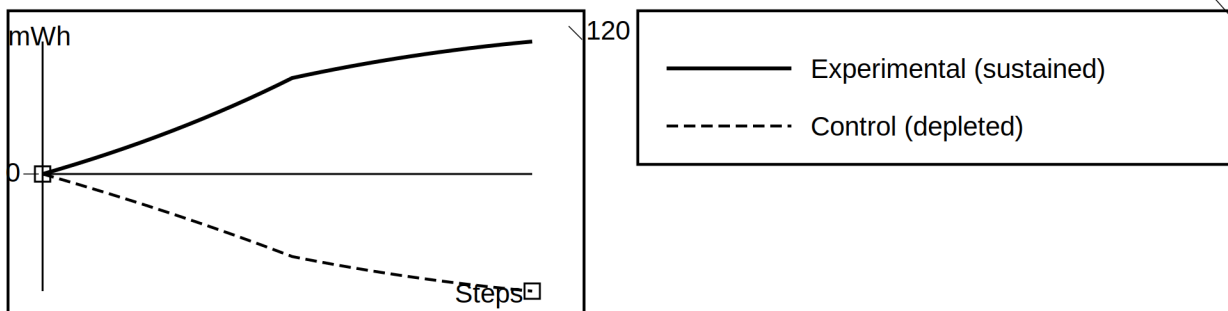
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Claim	Criterion	Measured	Threshold	Result
Phase 7 : Control Baseline				
A	Continuity	2000 pts	2000 pts	PASS
B	Boundedness	finite	finite	PASS
C	Robustness	std = 24.8	std <= 45	PASS
D	I/O gain	corr = 0.42	corr >= 0.2	PASS
E	Saturation	0.03	<= 0.20	PASS
Phase 8 : Synaptic Plasticity (STDP)				
F8.1	Weight entropy	0.94 bit drop	decrease	PASS
F8.2	STDP MI ratio	6.52x	> 1.20x	PASS
F8.3	Convergence	0.00%	< 10%	PASS
Phase 9 : Predictive Processing				
F9.1	Error reduction	63.9%	> 50%	PASS
F9.2	Random trend	p = 0.69	p > 0.05	PASS
F9.3	Adaptation	1.51x spike	> 1.3x + recover	PASS
Phase 10 : Multi-Modal Integration				
F10.1	Sync difference	0.70	> 0.20	PASS
F10.2	Weight entropy	2.31 to 0.0	decrease	PASS
F10.3	Convergence	0.18	< 0.20	PASS

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Configuration	Energy at 2000 Steps	Status
Control (EnergyConfig)	-4,697 mWh	Depleted
Experimental (BalancedConfig)	100.0 mWh	Sustained

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14 / 14 CLAIMS PASS : Phases 7 - 10 Validated

FIG. 7

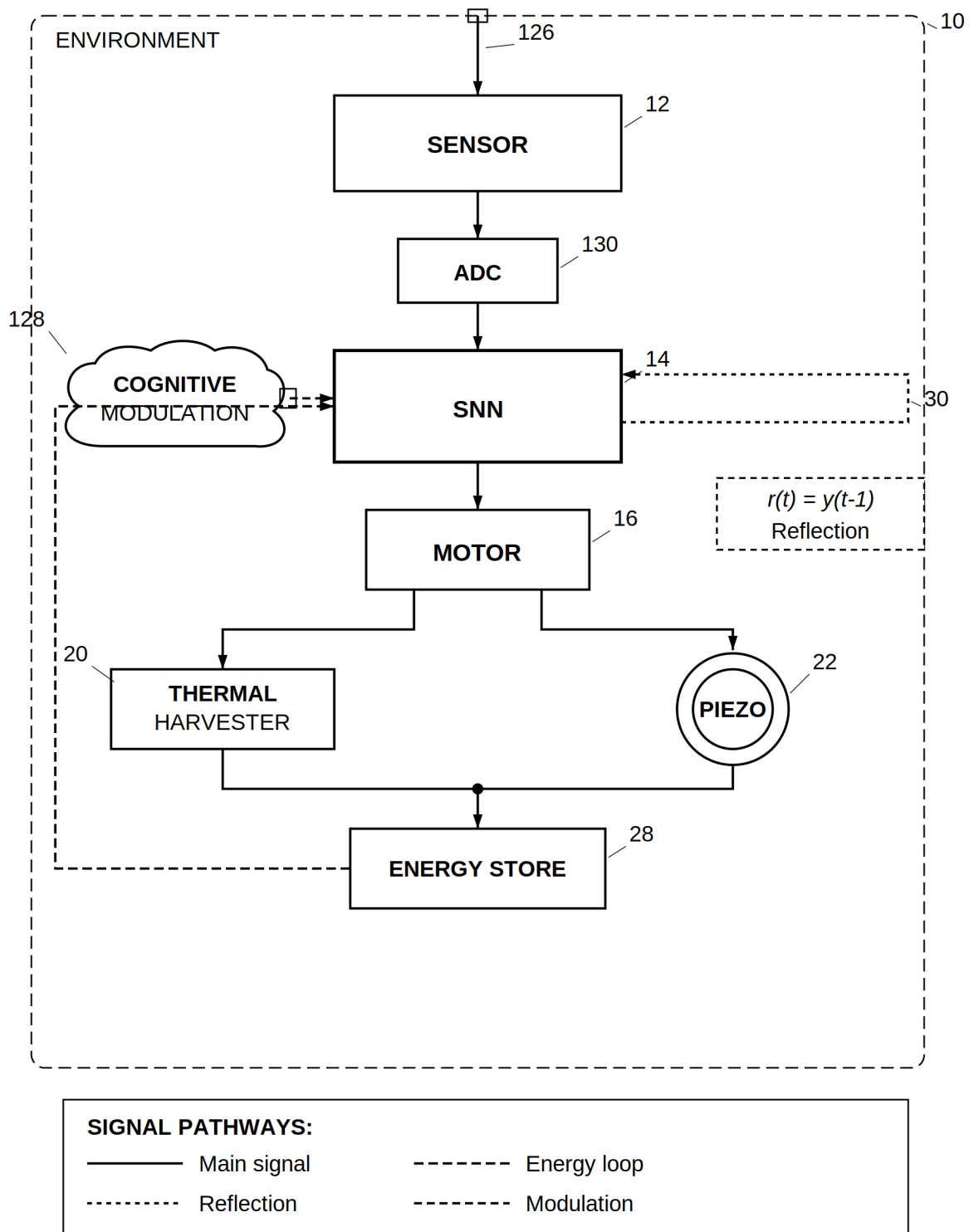
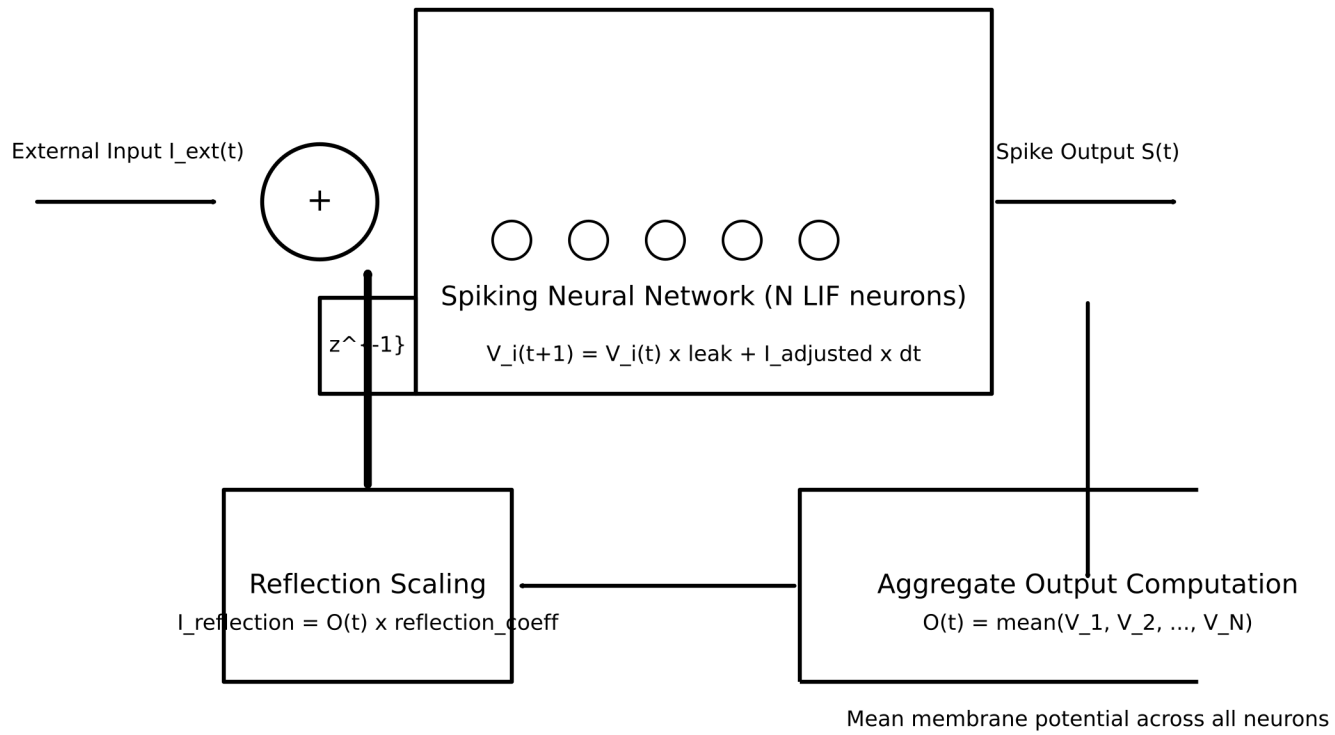


FIG. 8

PATENT B

Recursive Self-Observation Method and System for Spiking Neural Networks with Dynamic Reflection Coefficient

6 drawing sheets



Self-observation: the network observes its own prior aggregate state

$$I_{\text{adjusted}}(t+1) = I_{\text{ext}}(t+1) + I_{\text{reflection}}(t)$$

FIG. 1

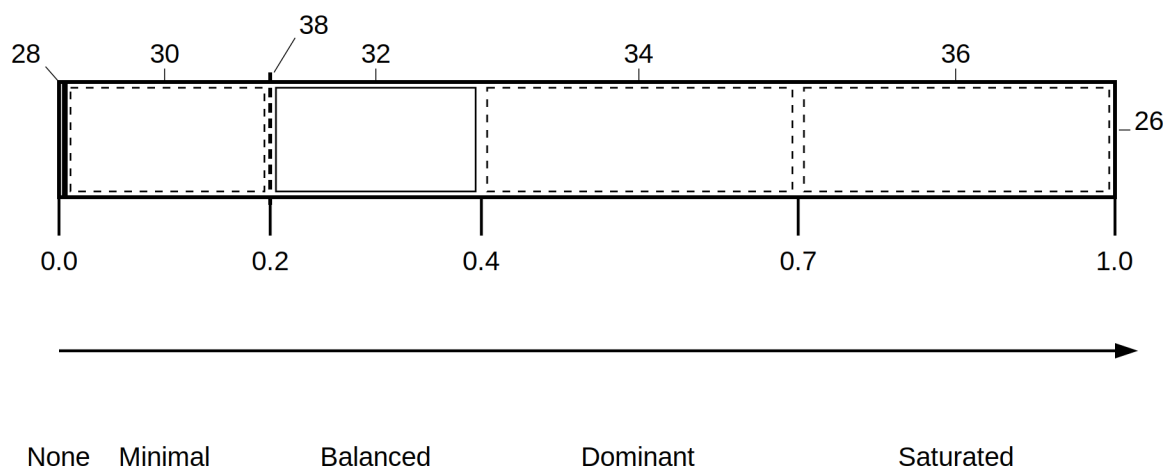
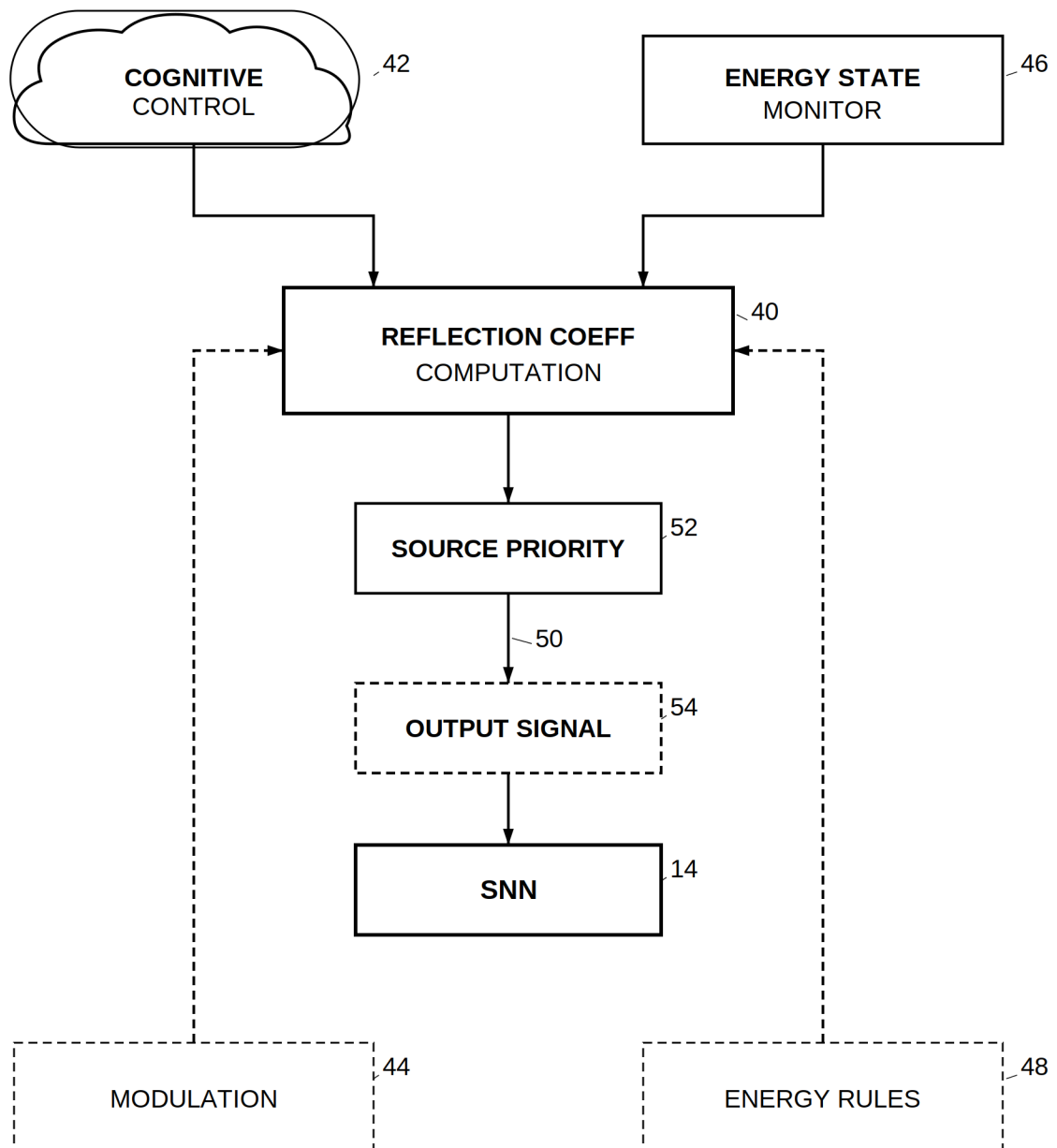


FIG. 2

**FIG. 3**

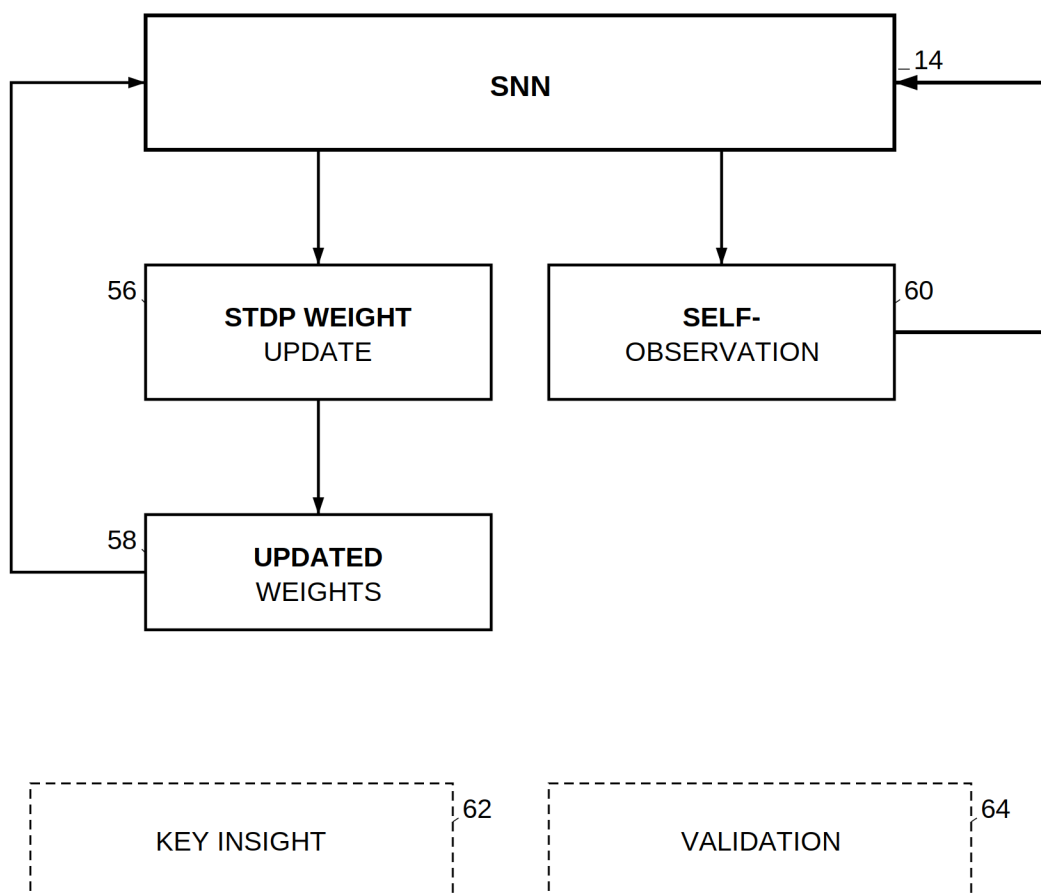


FIG. 4

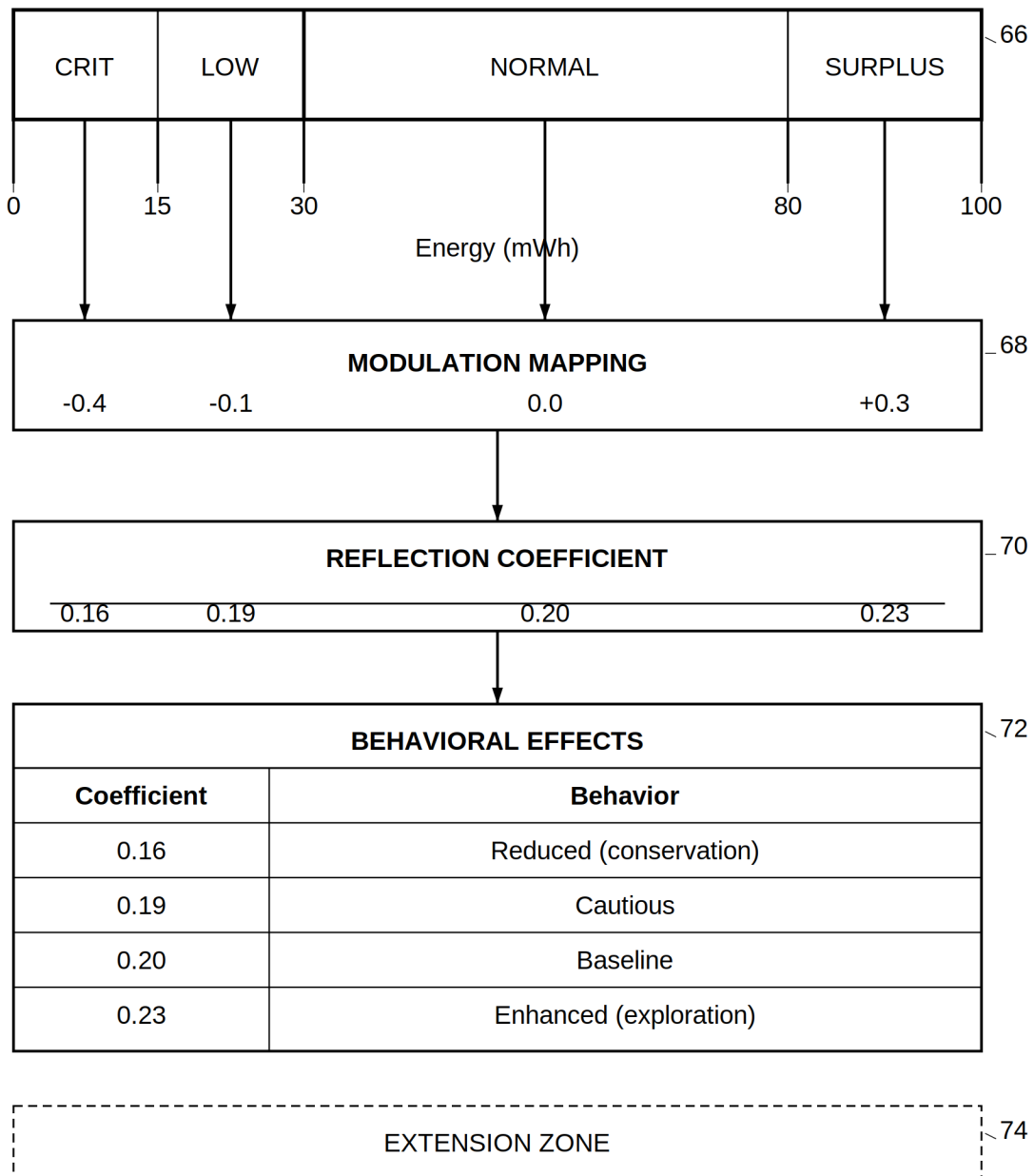
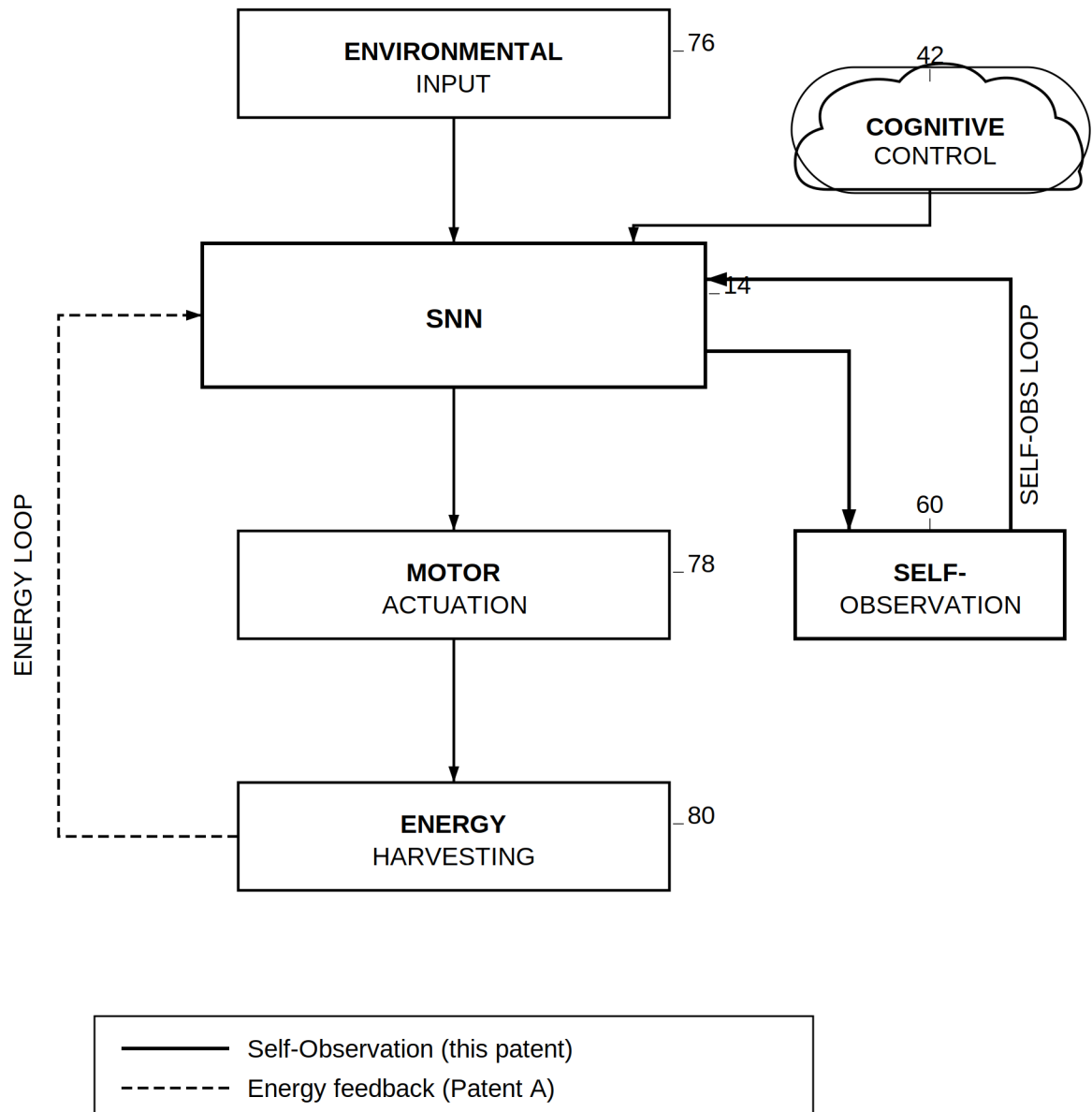


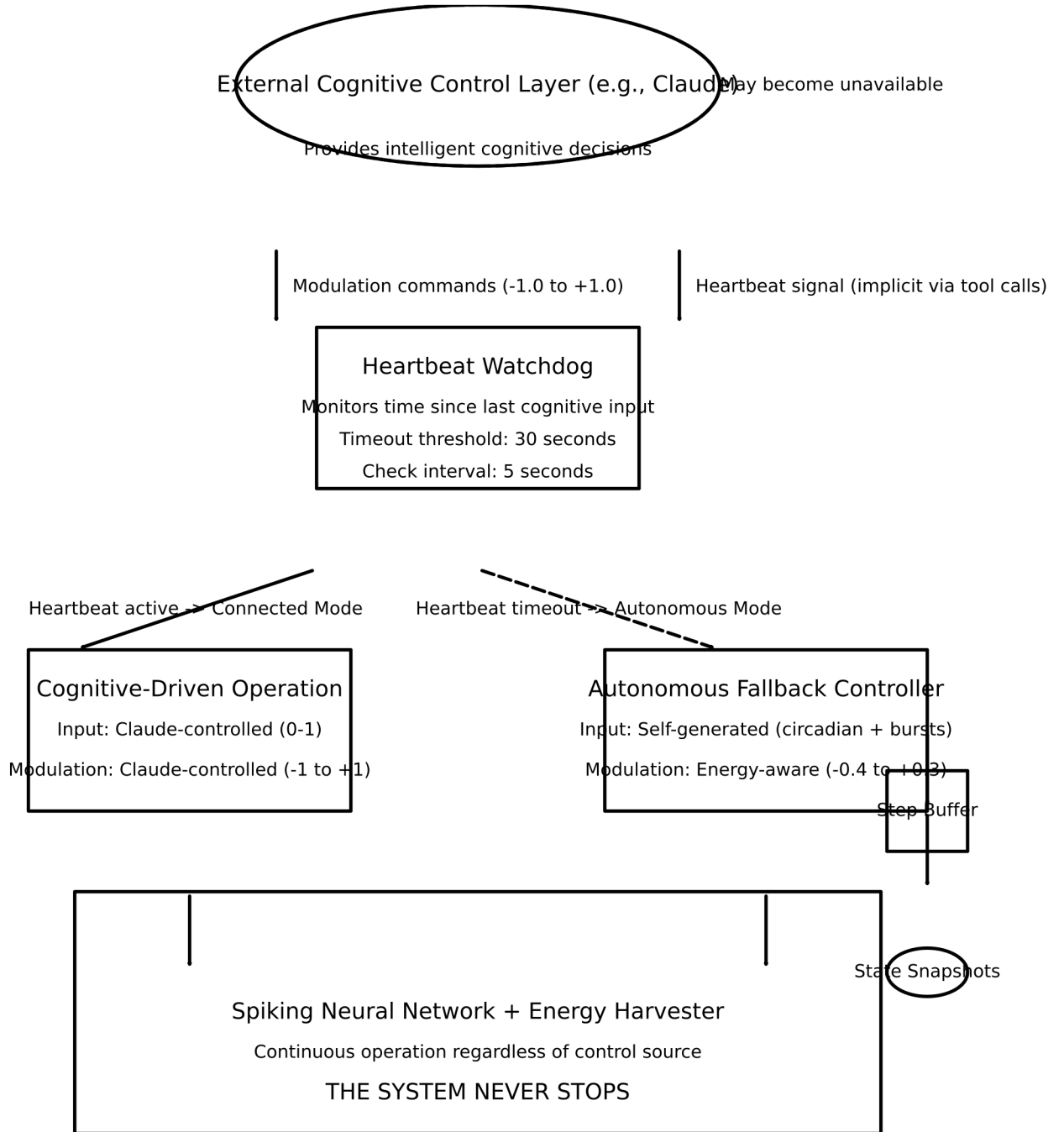
FIG. 5

**FIG. 6**

PATENT C

Autonomous Self-Regulation■and Cognitive Resynchronization in Neural Processing■Systems During Disconnection from

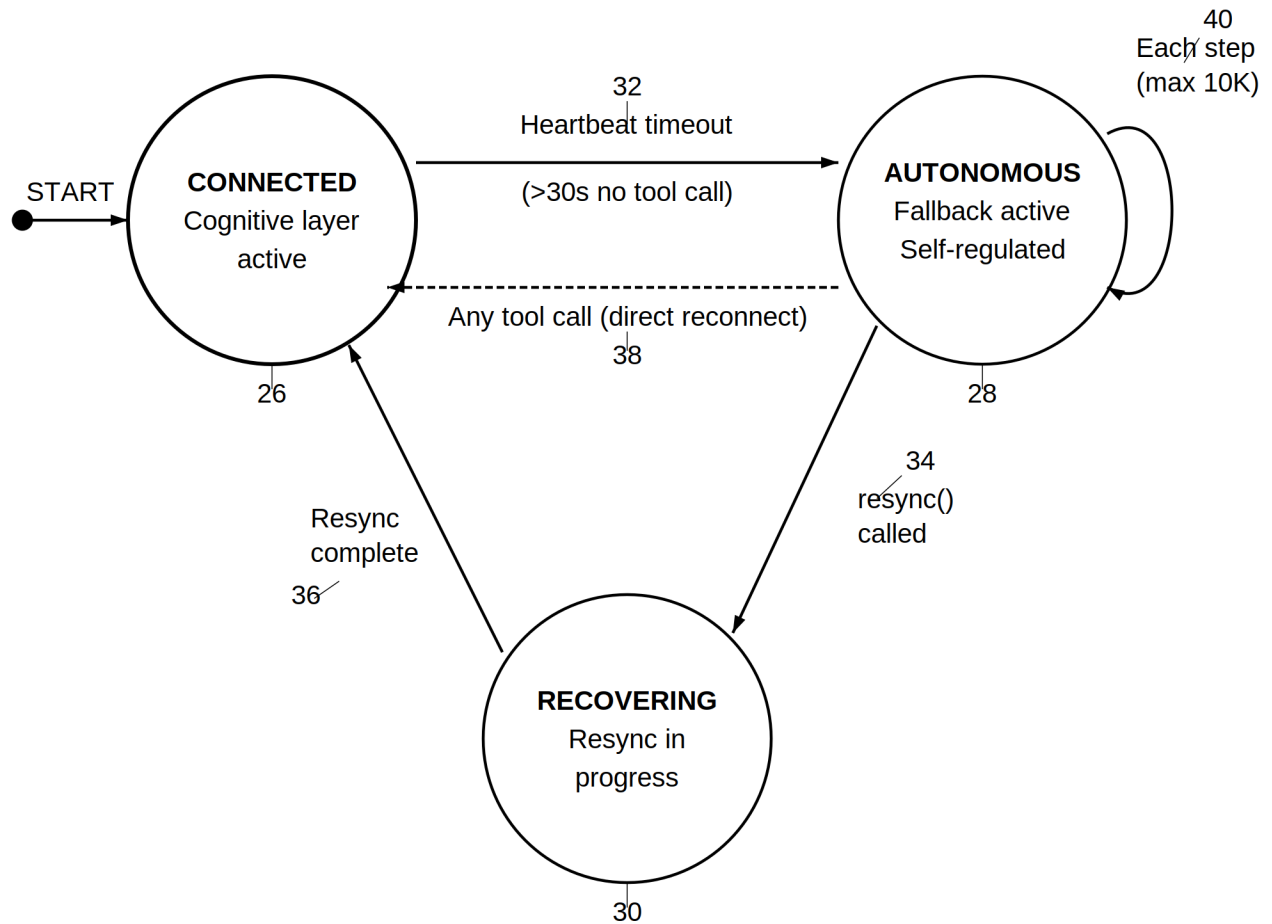
7 drawing sheets



Seamless transition: SNN continues processing without interruption

FIG. 1

The control source changes; the computation continues



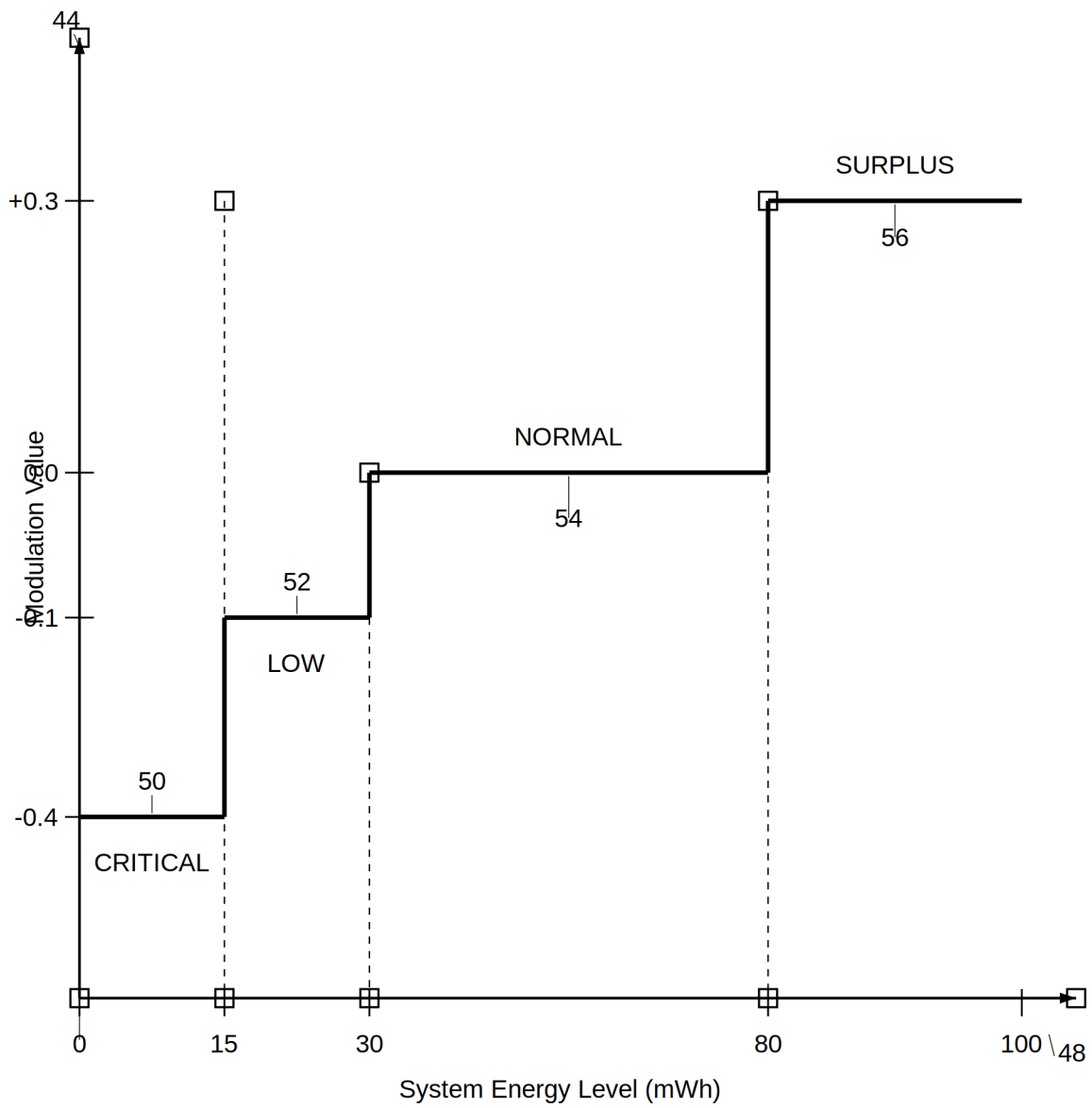
State Transition Legend:

26 **CONNECTED**: Cognitive layer provides input/modulation

28 **AUTONOMOUS**: Self-generated input, energy-aware modulation

30 **RECOVERING**: Buffer transmitted, restoring control

FIG. 2



Behavioral Effects:

CRITICAL (-0.4): Maximum conservation - inhibit neural activity

LOW (-0.1): Cautious operation - slightly reduce activity

NORMAL (0.0): Standard autonomous processing

SURPLUS (+0.3): Opportunistic exploration - utilize excess

FIG. 3

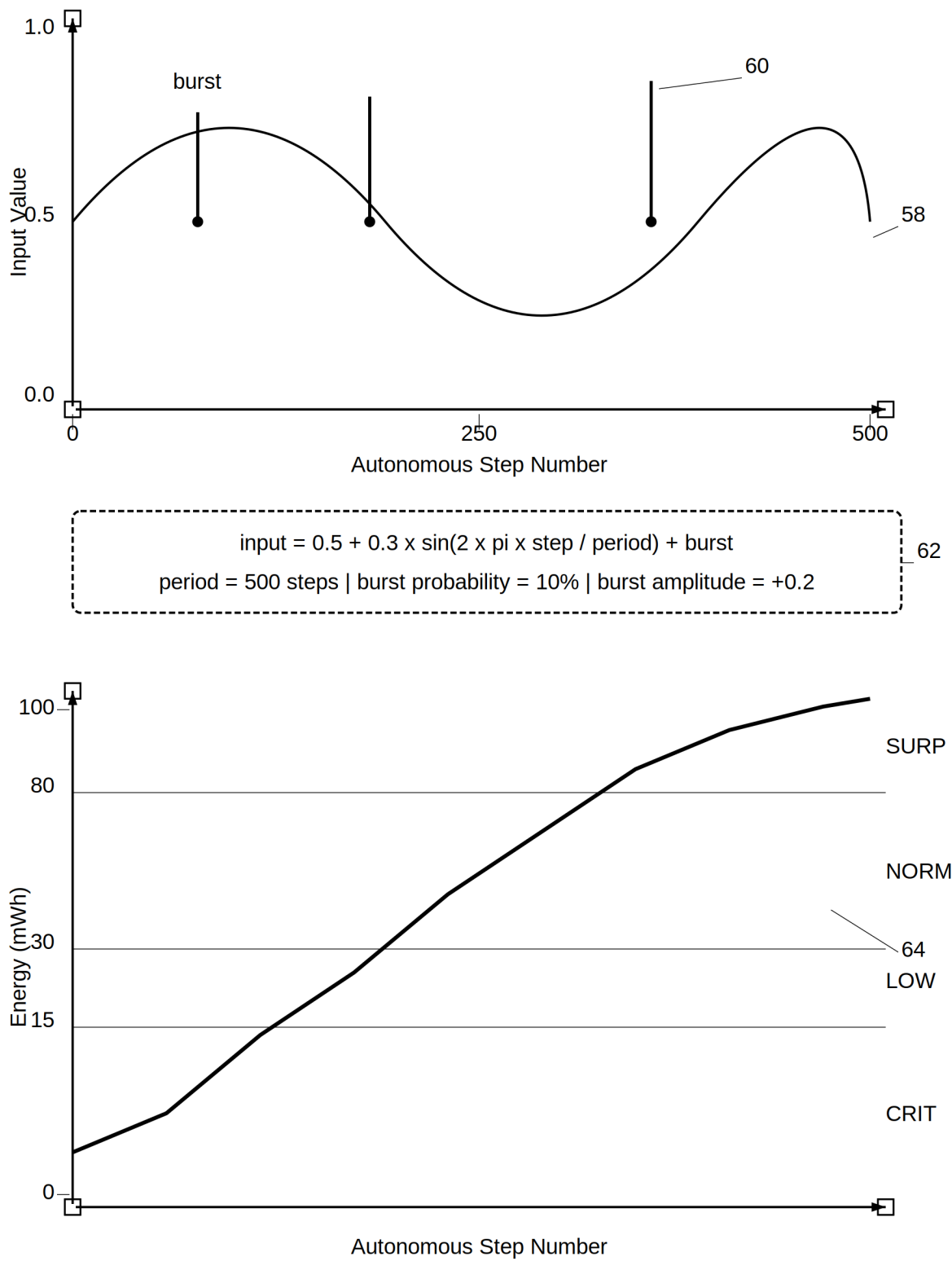
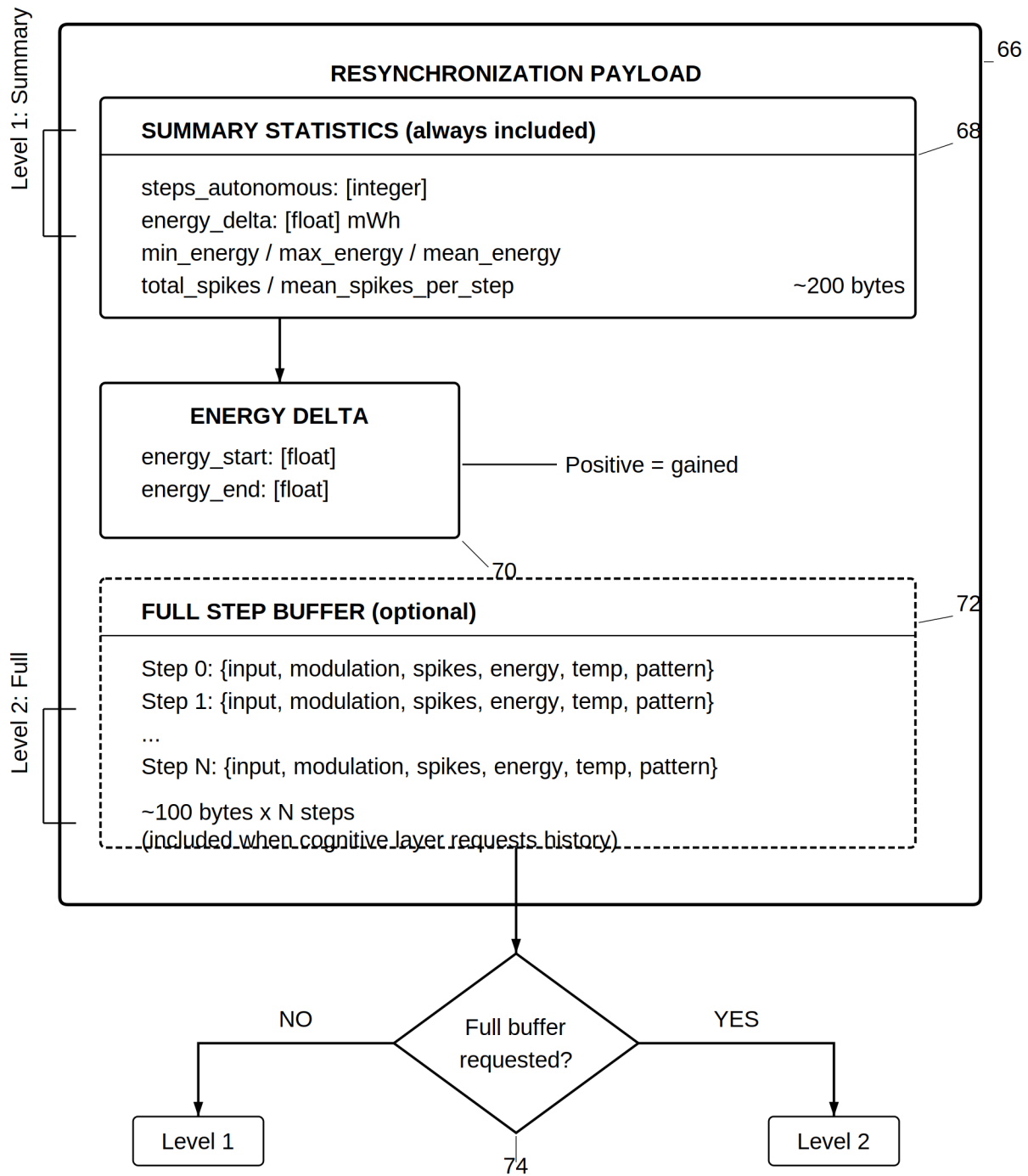
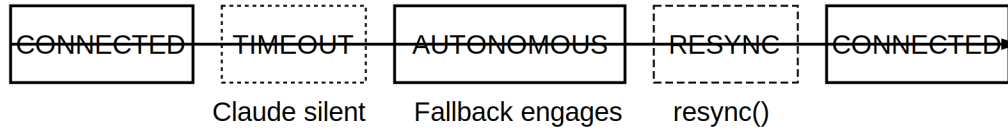


FIG. 4

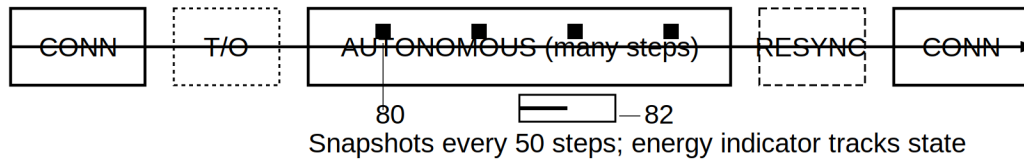
**FIG. 5**

NORMAL RECONNECTION

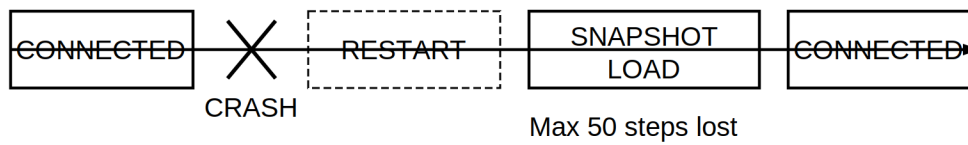
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**EXTENDED DISCONNECTION**

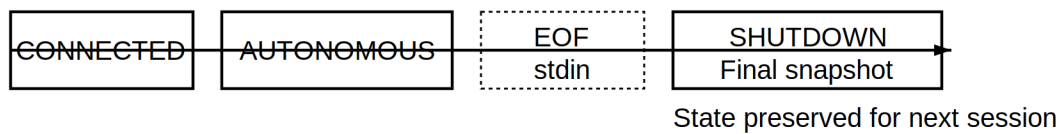
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**CRASH RECOVERY**

___ 84

**CLEAN SHUTDOWN**

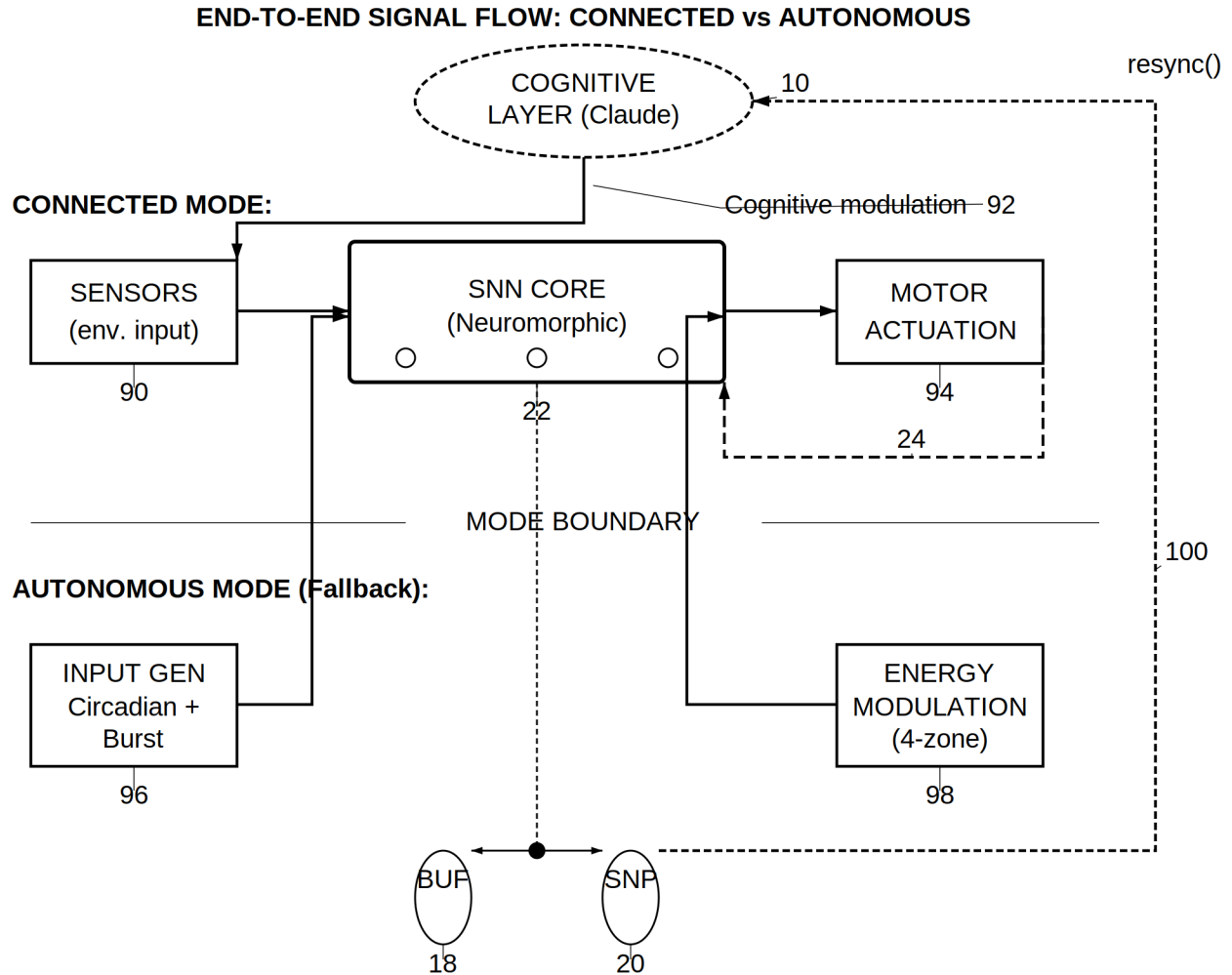
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**RECOVERY GUARANTEES**

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- Solid line: System actively processing
- - - - Dashed line: System in transition
- Bold markers: No data loss at state change

FIG. 6

**MODE SELECTION LOGIC:**

CONNECTED: Input from sensors 90, modulation from Claude 92

AUTONOMOUS: Input from generator 96, modulation from energy 98

Buffer 18 and snapshot 20 active during autonomous mode

Feedback path 100 transmits resync payload on reconnection

FIG. 7